School of Computing

TestSplit

TestSplit optimises CI pipelines by analysing historical test runtimes and distributing tests across parallel jobs using LPT and MULTIFIT scheduling, achieving a 2.31× speed-up (57% reduction in pipeline time).*

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1 THE PROBLEM

CI pipelines run tests one by one.

The bigger your project, the longer you wait.

Total Pipeline Time

Sequential
5:27 mins.

Parallel
2:26 mins.

* Evaluated on the apache/commons-lang GitHub repository

2 HOW IT WORKS

XML

Parse Test Reports

Reads JUnit XML output from your Maven Build

Smart Test Distribution

Group tests into equal-time parallel jobs to minimise total pipeline time.

Generate CI Config

Outputs a ready-to-use GitHub Actions or GitLab CI YAML File

3 PERFORMANCE IMPACT

TOTAL TESTS	SEQ. DURATION
65904	361.2s
across 42 profiling runs	unparallelised total

Before (Sequential Execution)

↓ 57% reduction in pipeline time (2.31× speed-up)

MAKESPAN	SPEED-UP
156.27s	2.31×
critical path · 8 jobs	13147 unstable · 26 outliers

After (Parallel Execution with TestSplit)

```yaml
jobs:
  build:
    run: mvn clean install -DskipTests

  test-job-1:
    run: mvn test -Dtest=ExtendedMessageFormatTest,FastDateParserTest

  test-job-2:
    run: mvn test -Dtest=StringUtilsTest,ArrayUtilsTest
```

Example generated job split

4 RESULTS

- 65,904 tests profiled (apache/commons-lang)
- 57% reduction in CI pipeline time
- 2.31× speed-up via parallel execution
- Validated on real-world open-source CI pipelines
- No modification to existing test suites required

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Docker · Node.js · TypeScript · React w/ Vite · GitLab · GitHub

Validated across multiple repositories
70,000+ tests